

CSA6102 – DIGITAL FORENSICS AND CYBERCRIME INVESTIGATION

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EXP 2

1. Write a Python program to verify the chain of custody of digital evidence using file hashing



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# =====
# Chain of Custody Verification using File Hashing (SHA-256)
# Google Colab Compatible
# =====

import hashlib

# -----
# Function to calculate SHA-256 hash of a file
# -----
def calculate_hash(filename):
    sha256 = hashlib.sha256()

    with open(filename, "rb") as file:
        while True:
            data = file.read(4096)
            if not data:
                break
            sha256.update(data)

    return sha256.hexdigest()

# -----
# Step 1: Create a digital evidence file
# -----
filename = "digital_evidence.txt"

with open(filename, "w") as file:
    file.write("Digital Evidence\n")
    file.write("Case ID: 101\n")
    file.write("Collected by: Investigator A\n")

print("Evidence file created successfully.\n")

# -----
# Step 2: Generate the original hash
# -----
original_hash = calculate_hash(filename)

print("==== ORIGINAL HASH =====")
print(original_hash)

# -----
# Step 3: Verify the file (Before Tampering)
# -----
print("\n===== VERIFYING CHAIN OF CUSTODY =====")

current_hash = calculate_hash(filename)
```

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❏ ❶ if current_hash == original_hash:
    print("Verification Successful")
    print("Chain of Custody Maintained")
else:
    print("Verification Failed")
    print("Evidence Integrity Compromised")

# -----
# Step 4: Simulate Tampering
# -----
print("\nSimulating Unauthorized Modification...\n")

with open(filename, "a") as file:
    file.write("Tampered Data Added\n")

# -----
# Step 5: Verify Again
# -----
tampered_hash = calculate_hash(filename)

print("===== NEW HASH =====")
print(tampered_hash)

print("\n===== VERIFYING AFTER TAMPERING =====")

if tampered_hash == original_hash:
    print("Verification Successful")
    print("Chain of Custody Maintained")
else:
    print("Verification Failed")
    print("Evidence Has Been Modified")
    print("Chain of Custody Broken")

# -----
# Step 6: Display File Contents
# -----
print("\n===== FILE CONTENT =====")

with open(filename, "r") as file:
    print(file.read())

# -----
# Step 7: Conclusion
# -----
print("===== CONCLUSION =====")
print("Chain of Custody ensures that digital evidence")
print("remains unchanged from collection to presentation.")
print("SHA-256 hashing helps detect any unauthorized")
print("modification of the evidence.")

```

OUTPUT :

```
*** Evidence file created successfully.

===== ORIGINAL HASH =====
42d44b6428814a7b37333576ec831adede47f67b06e420db5b323cc3875280dc

===== VERIFYING CHAIN OF CUSTODY =====
Verification Successful
Chain of Custody Maintained

Simulating Unauthorized Modification...

===== NEW HASH =====
2292136436cb732b6ad2fe87486425b4051e7b097b174be2f99eaf9fd1f1872d

===== VERIFYING AFTER TAMPERING =====
Verification Failed
Evidence Has Been Modified
Chain of Custody Broken

===== FILE CONTENT =====
Digital Evidence
Case ID: 101
Collected by: Investigator A
Tampered Data Added

===== CONCLUSION =====
Chain of Custody ensures that digital evidence
remains unchanged from collection to presentation.
SHA-256 hashing helps detect any unauthorized
modification of the evidence.
```